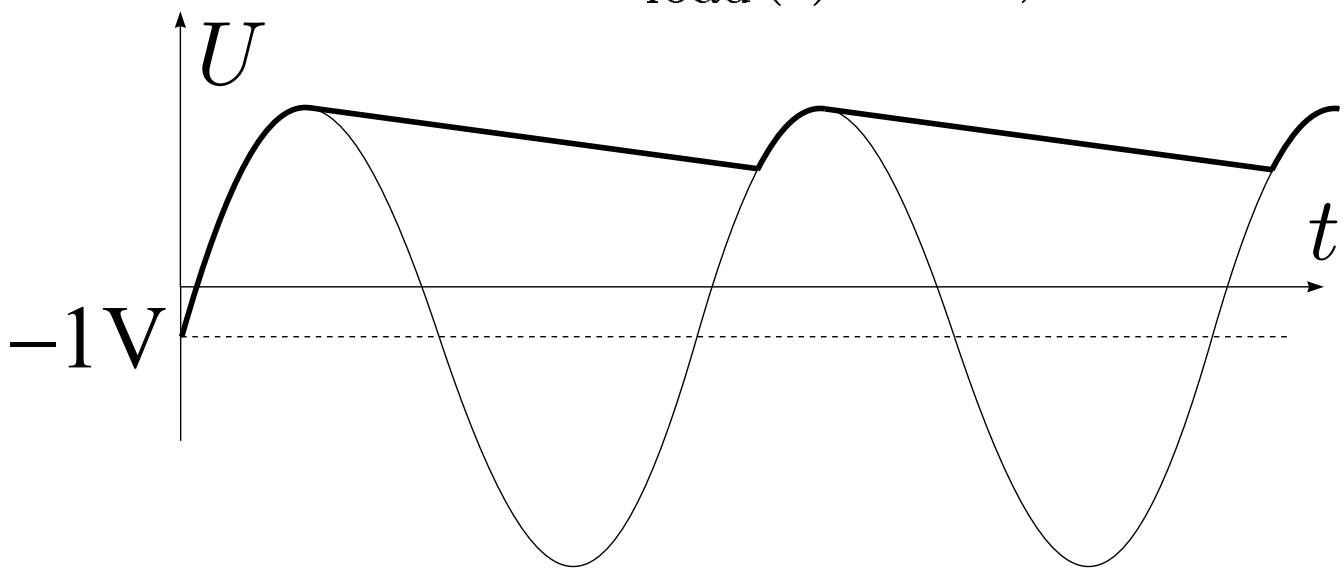
6. **Rectifier** (8 points)

1) Since none of the DC current through the load can come from the capacitor, all must come through the diode. Hence, the average current through the diode is also $I = 2\,\mathrm{mA}$, and the average power dissipation is obtained by multiplying it with the diode voltage $u = 1\,\mathrm{V}$: $P = 2\,\mathrm{mW}$.

2) If the diode is open, $U_{\mathrm{load}}(t) = U_0 \cos(\omega t) - u$. If the diode is closed [i.e. $U_0 \cos(\omega t) < U_{\mathrm{load}}(t) + u$], the capacitor discharges through the load. However, the relative change of the

voltage of the capacitor has to be small (otherwise $\Delta I/I$ would not be small). The respective load voltage as a function of time is sketched in the Figure. So, we can use the above written Kirchoff's law with $U_{\rm load}(t) \approx IR$, hence $U_0 = IR + u = 21\,\mathrm{V}$.



3) The change of the voltage of the capacitor during the discharge cycle can be estimated as $\Delta U = \Delta Q/C$, where the capacitor's charge drop $\Delta Q = It$, and t is the discharge time. Since the discharge cycle occupies almost all the period (see Figure), we can use $t \approx 1/\nu$. Further, $\Delta I/I = \Delta U/U = \Delta Q/CU = \Delta Q/CIR = 1/CR\nu$. Hence, $C \geq 100/R\nu = 200\,\mu\mathrm{F}$.

4) Initially, the capacitor is empty, so that the charge flowing through the capacitor during the first cycle is $Q = CIR$. Hence, the average power $P_1 = Qu\nu = CIRu\nu = 200\,\mathrm{mW}$.